

SkateGuard Project Progress Report

Weeks 5 and 6

1. Report Scope

The final development phase connected the embedded system to a complete rider interface and then exercised the integrated firmware, wireless link, GPS, dashboard and session storage. The emphasis was on system behavior rather than adding another separate application.

2. Self Hosted Dashboard Architecture

The board itself hosts the dashboard through a WebServer on port 80. The rider's phone and board remain on the same phone hotspot, so dashboard telemetry and commands use local HTTP requests. The page polls the live telemetry endpoint twice per second. The design avoids a separate companion application and keeps the telemetry path local.

The board does the work; the phone and the wrist only show it

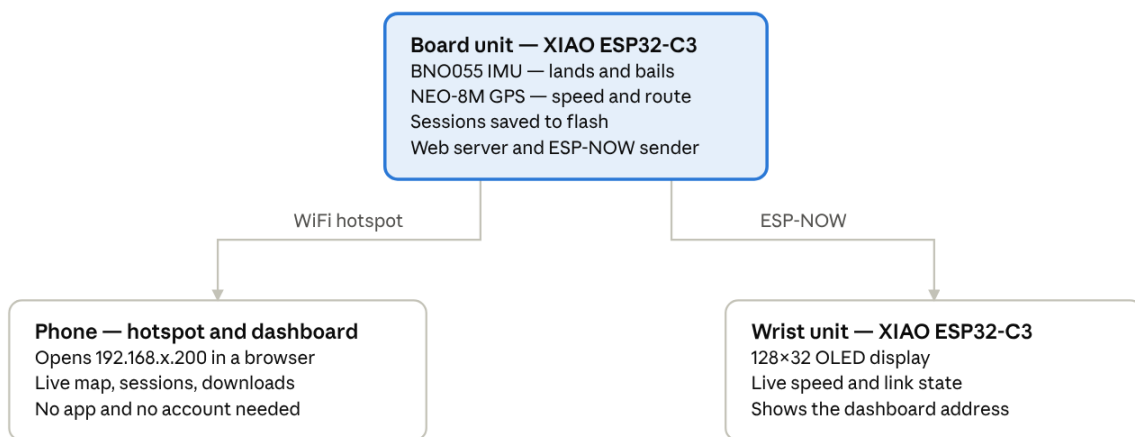


Figure 1. SkateGuard system architecture: board unit, phone dashboard and wrist unit.

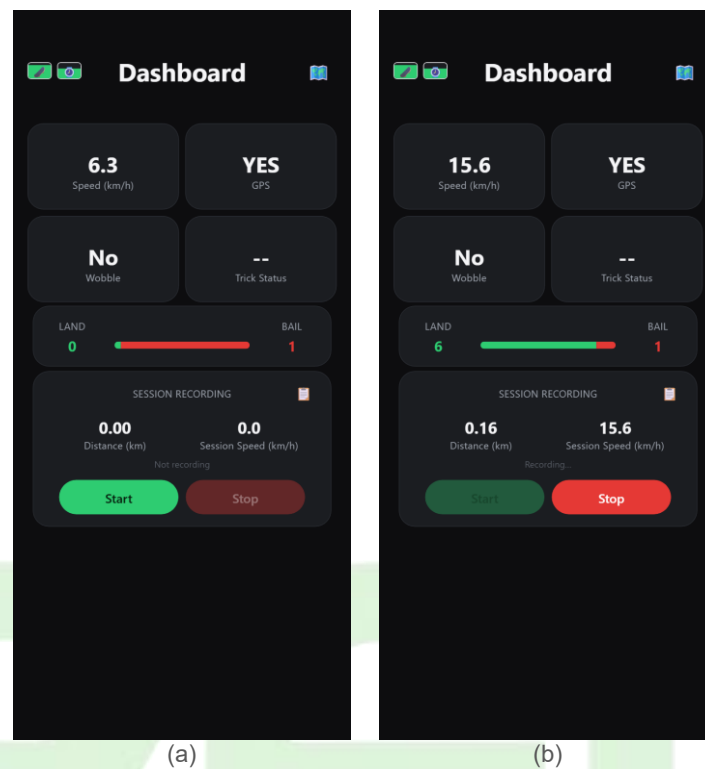


Figure 2. Board hosted dashboard on the phone: (a) live telemetry before recording, (b) session recording in progress.

2.1 Dashboard and Storage Parameters

Parameter	Value	Purpose / implementation
Web server	ESP32 WebServer, port 80	Board hosted user interface
Live endpoint	GET /api/live	Live telemetry JSON
Debug endpoint	GET /api/debug	Motion, GPS, radio and memory diagnostics
Sessions endpoint	GET /api/sessions	Saved session data
Command endpoint	/api/command?cmd=	Start, stop, save, discard and delete
Motion log	/log and /clear	Download or clear stage log
Dashboard polling	500 ms	Two live updates per second
Saved sessions	Maximum 7	FIFO session storage
Route sampling	30 s	Maximum 20 stored route points per session
Manifest	/manifest.json	Add to Home Screen support

3. Session Recording

A ride is recorded locally and is not written as a saved session until the rider selects Save. The post ride record contains distance, average speed, top speed, land count, bail count and wobble count. GPS route points are stored separately and rendered on the dashboard. The system also supports sessions without GPS, which was important for testing the dashboard behavior.

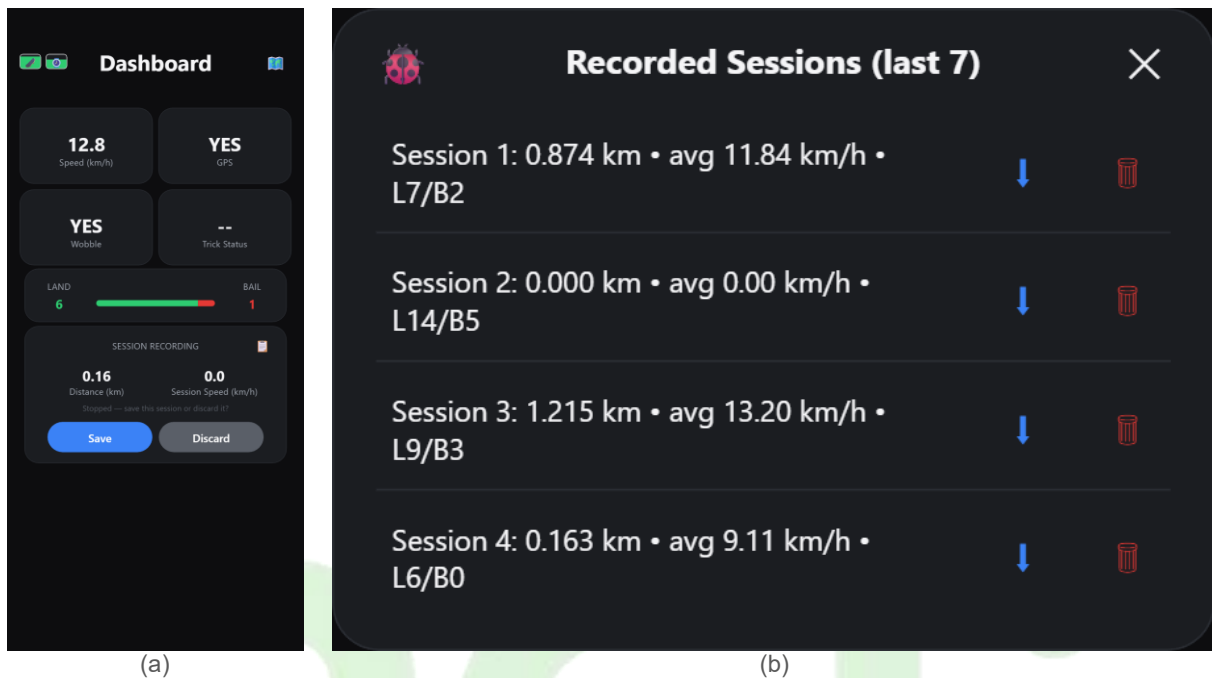


Figure 3. Session recording: (a) Save or Discard after stopping a ride, (b) recorded sessions list (last 7).

4. Stage Logging

The motion stage log is written at 500 ms intervals. The firmware stores stage, gyro Y, accumulated rotation, land confirmation swing, event, kickflip and bail counters, active and absolute motion windows, cooldown window, oscillation count and the other gyro axes. The log is capped at 2000 rows.

Debug		×
Stage	IDLE	
Gyro Y	-0.9 deg/s	
Accum Rotation	0.0 deg	
Land Confirm Swing	0.0 deg	
Active Window	0 / 2000 ms	
Absolute Window	0 / 4000 ms	
Cooldown Window	0 / 500 ms	
Oscillations	0 / 3	
Loop Max / Avg	12 / 3.8 ms	
Land Count	1	
Bail Count	0	
Battery	86% (4.02 V)	
GPS Satellites	9	
GPS HDOP	0.9	
GPS Fix Age	0.4 s	
GPS Data In	9426 chars	
GPS Checksum Fails	0	
WiFi	ch 6 -54 dBm	
ESP-NOW	wearable linked	
Free Heap	175 KB	
Uptime	9s	
Download stage_log.csv Clear log		

Figure 4. Debug page showing motion stage, timing windows, counters, GPS, radio and memory diagnostics with stage log download.

4.1 Logging Parameters

Parameter	Value	Purpose / implementation
Stage log interval	500 ms	Periodic motion state record
Stage log capacity	2000 rows	Maximum stored motion log
Session capacity	7 rides	Most recent saved rides retained
Route capacity	20 points/session	Maximum route points
Route sample interval	30 s	Distance between stored route points

5. Integrated Testing

Testing covered motion logging, loop health, GPS status, the ESP NOW radio link, dashboard workflows and field route rendering. Radio testing included changing power up order, moving out of range and returning, and changing the phone hotspot so that the channel moved. Dashboard testing exercised Save, Discard, the seven session limit, deletion and downloads with and without GPS.

5.1 Testing Results Recorded in the Project Report

Parameter	Value	Purpose / implementation
Trick detection	stage_log.csv, threshold and event monitoring	Thresholds recorded and log download verified
Loop health	Loop period shown on debug page	Instrumentation added to identify missed motion events
GPS	Outdoor and indoor fix tests	Outdoor fix acquired with 4+ satellites; no indoor fix
Radio link	Power order, range and channel changes	Wrist re found board without reboot
Dashboard	Save, discard, cap, delete and download	Paths exercised; no GPS route alignment issue reproduced and fixed
Field use	Real rides and route display	Routes rendered over OpenStreetMap tiles

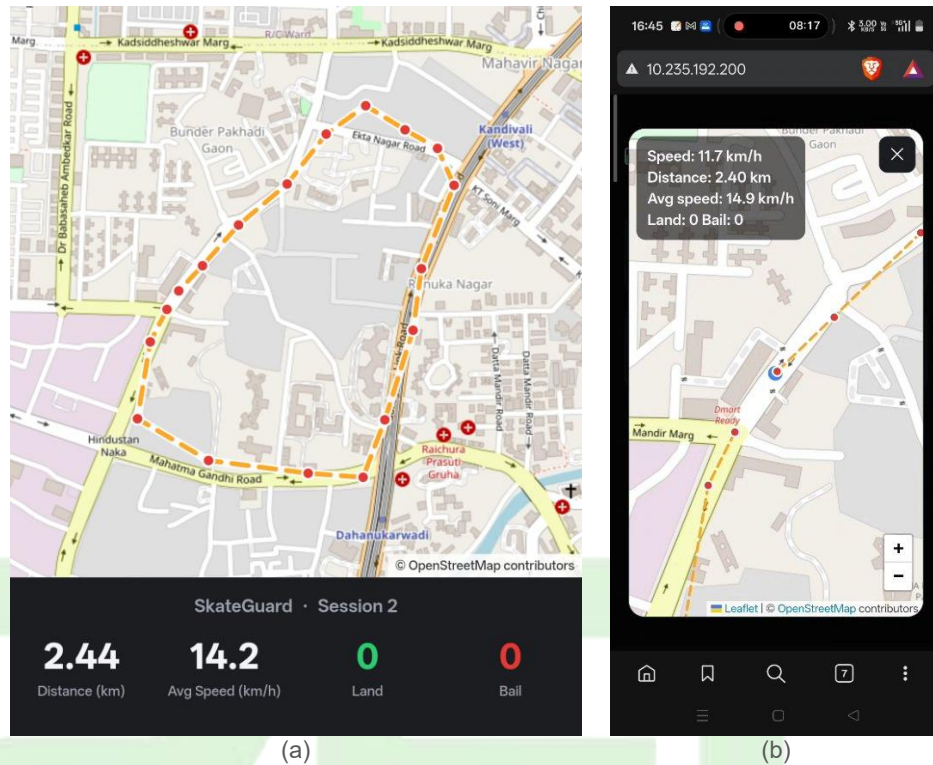


Figure 5. Field rides rendered over OpenStreetMap tiles: (a) saved session route with ride summary, (b) live map view on the phone.

6. Dashboard Issue and Correction

During dashboard testing, a route alignment problem was reproduced when a no GPS session was stored between GPS recorded sessions. The session handling was corrected so that route information remains associated with the correct ride instead of being shifted by session position.

7. Network Test Dataset

7.1 WiFi and ESP NOW Test Data

Sample	Timestamp ms	Wifi connected	Wifi rssi dbm	Hotspot channel	Espnow ready	Packets sent	Packets lost	Packets received	Packet loss percent	Dashboard response ms
1	0	YES	-55.300	6	YES	1	0	1	0.000	5.300
2	100	YES	-56.700	6	YES	2	0	2	0.000	8.900
3	200	YES	-52.500	6	YES	3	0	3	0.000	7.900
4	300	YES	-53.900	6	YES	4	0	4	0.000	4.800
5	400	YES	-51.900	6	YES	5	0	5	0.000	10.300
6	500	YES	-54.700	6	YES	6	0	6	0.000	6.500
7	600	YES	-53.000	6	YES	7	0	7	0.000	6.400
8	700	YES	-49.600	6	YES	8	0	8	0.000	8.400
9	800	YES	-51.800	6	YES	9	0	9	0.000	10.300
10	900	YES	-46.100	6	YES	10	0	10	0.000	6.000

Table shows the first 10 rows from the test dataset for traceability. The complete CSV is retained separately.

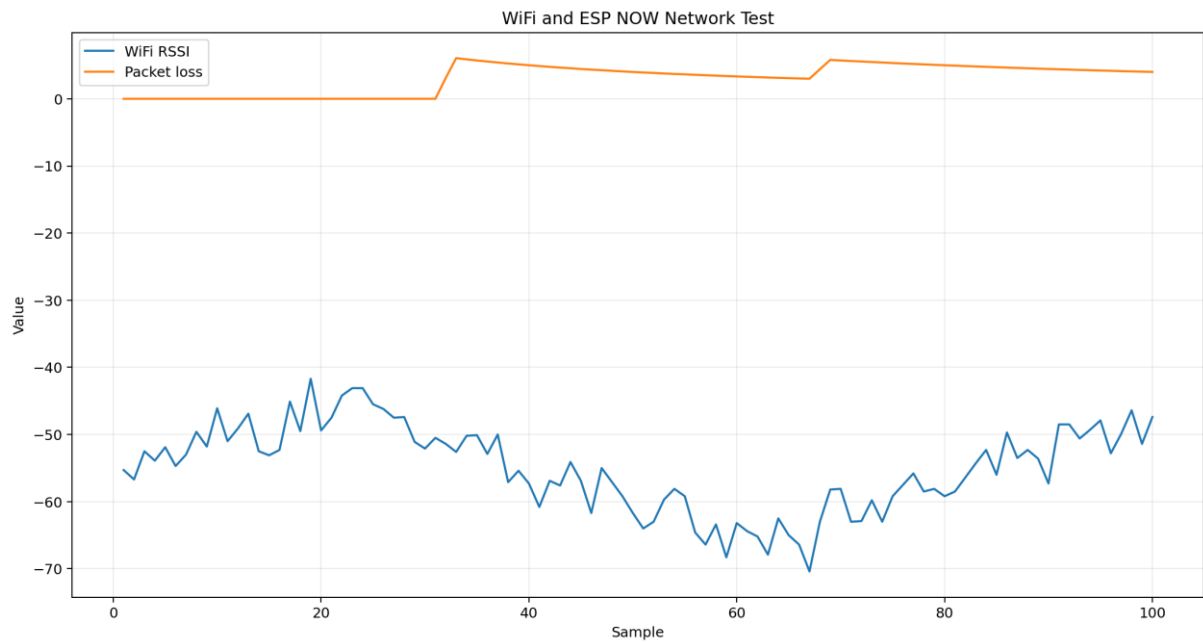


Figure 6. Network validation visualization showing RSSI and packet loss behavior.

8. Battery Monitoring

8.1 Battery Monitoring Test Data

Sample	Timestamp ms	Adc voltage v	Battery voltage v	Battery percent raw	Battery percent reported	Battery status
1	0	0.999	4.159	95.400	95	NORMAL
2	100	1.005	4.185	98.400	100	NORMAL
3	200	0.998	4.158	95.400	95	NORMAL
4	300	1.002	4.173	97.000	95	NORMAL
5	400	1.002	4.172	96.900	95	NORMAL
6	500	0.999	4.162	95.800	95	NORMAL
7	600	0.994	4.141	93.500	95	NORMAL
8	700	0.996	4.147	94.100	95	NORMAL
9	800	0.994	4.138	93.100	95	NORMAL
10	900	0.990	4.124	91.500	90	NORMAL

Table shows the first 10 rows from the test dataset for traceability. The complete CSV is retained separately.

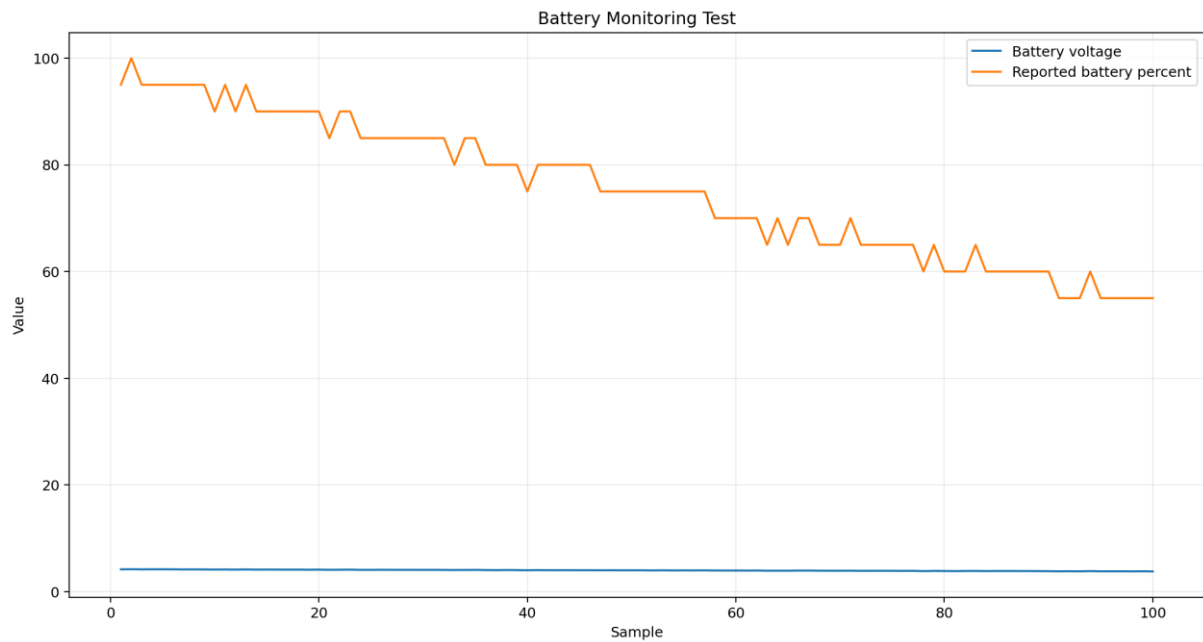


Figure 7. Battery voltage and reported percentage test visualization.

9. Integrated System Dataset

9.1 Integrated System Test Data

Sample	Timestamp ms	Speed kmh	Satellites	Gps fix	Gyro y dps	Tilt angle z deg	Motion stage	Wobble active	Kickflip count	Bail count	Wifi rssi dbm	Espnow ready	Battery percent
1	0	7.160	7	YES	23.810	3.670	IDLE	0	0	0	-55.300	YES	95
2	100	9.280	8	YES	37.030	3.780	IDLE	0	0	0	-56.700	YES	100
3	200	10.120	7	YES	43.530	3.870	IDLE	0	0	0	-52.500	YES	95
4	300	9.160	6	YES	40.300	3.920	IDLE	0	0	0	-53.900	YES	95
5	400	9.570	8	YES	23.300	4.010	IDLE	0	0	0	-51.900	YES	95
6	500	11.210	9	YES	7.740	4.140	IDLE	0	0	0	-54.700	YES	95
7	600	10.960	8	YES	-1.890	4.360	IDLE	0	0	0	-53.000	YES	95
8	700	12.450	8	YES	-13.360	4.380	IDLE	0	0	0	-49.600	YES	95

Table shows the first 8 rows from the test dataset for traceability. The complete CSV is retained separately.

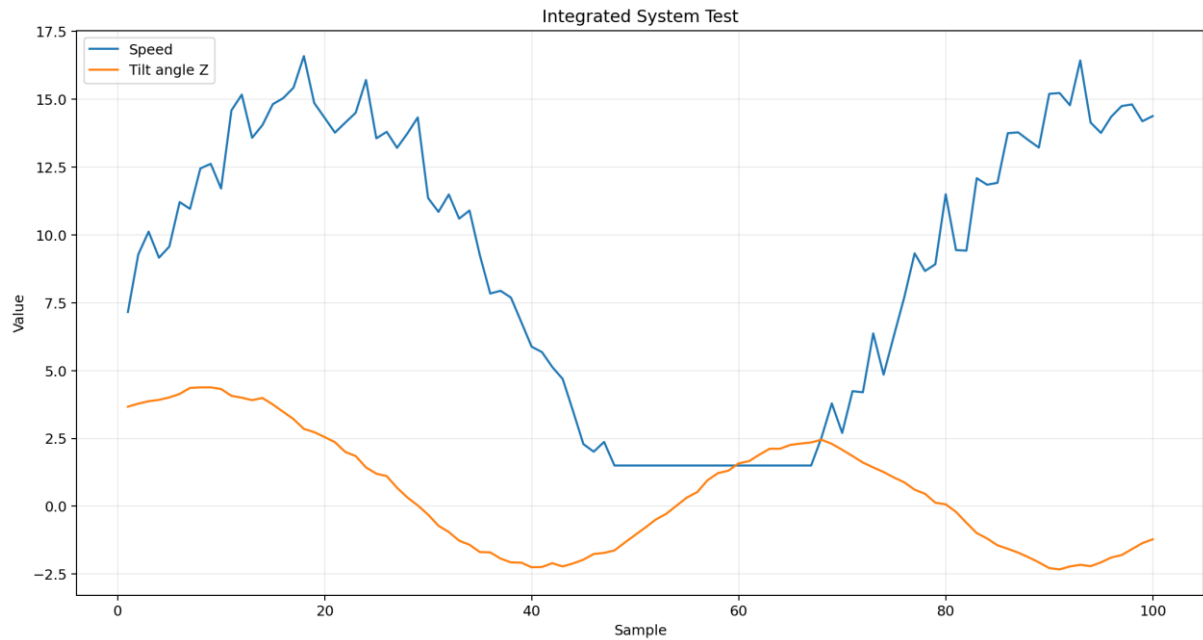


Figure 8. Integrated test visualization combining movement and system state variables.

10. Remaining Validation Work

- Confirm the kickflip axis with a hand counted real kickflip test.
- Compare automatic land and bail counts against the hand counted result.
- Confirm wobble detection and wrist motor response during a real riding speed wobble.
- Measure battery life over a complete session for both 1500 mAh units.
- Record enclosure print material, layer height, infill and finished dimensions.

11. Current System Status

The complete hardware, firmware and dashboard stack is assembled and running. Bench and browser validation are complete and initial field rides have been recorded. Final detection accuracy has not yet been quantified on real rides, so the remaining work is focused on validation and documentation rather than basic system construction.